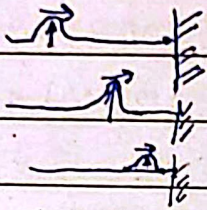


Wave

"is distance from equilibrium position".

Particles do not travel.



in this energy is travelling.

Mechanical $\rightarrow$ require medium

Non-mechanical $\rightarrow$ don't require medium

General eq of wave

Transverse [solid, liquid, surface]

"perpendicular to the wave motion"

e.g. - EM waves.

90° $\rightarrow$ motion

longitudinal $\rightarrow$ [solid, liquid, gas]

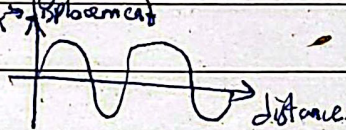
"parallel"

• As some quantity oscillate 'y'

$y \rightarrow$ displacement, pressure, density

y depends on x, t

$$y = f(x, t)$$



• Condition for wave motion

$$\frac{d^2 y}{dt^2} = k \frac{d^2 y}{dx^2} \quad (k \neq 0)$$

y should be defined for all values of x and t . like if it is $x = z/t \rightarrow$ so here we can put $t=0$ so.

(i) $y = A \sin \omega t$

$$\frac{dy}{dt} = \omega A \cos \omega t \rightarrow \frac{d^2 y}{dt^2} = -\omega^2 A \sin \omega t$$

$$\left[\frac{dy}{dx} = 0 \right] \text{ As NO } x \text{ variable}$$

[so Not a wave]

(ii) $y = A \sin(\omega t - kx)$

Proving eq. 1

$$\frac{dy}{dt} = \omega A \cos(\omega t - kx)$$

$$\frac{d^2 y}{dt^2} = -\omega^2 A \sin(\omega t - kx)$$

$$\frac{dy}{dx} = -kA \cos(\omega t - kx)$$

$$\frac{d^2 y}{dx^2} = -k^2 A \sin(\omega t - kx)$$

$$\frac{d^2 y}{dt^2} = \frac{d^2 y}{dx^2}$$

$$= \frac{\omega^2}{k^2} \rightarrow \frac{d^2 y}{dt^2} = \frac{\omega^2}{k^2} \frac{d^2 y}{dx^2}$$

Proves eq. 1

is constant square of wave velocity

$\rightarrow$ In this form general wave

$$y = f(ax + bt)$$

(a, b) constant

GENIUS

General eqs

Wave speed $\rightarrow$ energy $\rightarrow$ $ax+bt$
 $v = \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{b}{a}$

$\therefore$ when both have same sign $\{ ax+bt \} \rightarrow -ax-bt$ } travelling in -ve x direction

$\therefore$ To find wave

velocity $= v = \frac{bt}{ax}$

To find amplitude $\rightarrow$ it's max displacement

$y = \frac{10}{5+(x-2b)^2}$ To find max $\rightarrow$ min should be 0. so it will be 0.

eq of harmonic $\rightarrow A \sin(\omega t - \frac{\omega}{v} x + \phi)$

$k = \frac{2\pi}{\lambda}$
 $v = f\lambda$ wave no λ

Q) $y = 0.02 \sin\left(\frac{x}{0.01} + \frac{t}{0.05}\right) \rightarrow y = A \sin(\omega t - kx + \phi)$ } moving in negative direction

$y = A \sin(\omega t - kx + \phi)$ } moving +ve direction
 phase Initial phase at origin when $x=0, t=0$

(a) Amplitude = 0.02

(b) $A(x) = k = \frac{2\pi}{\lambda}$

$\frac{1}{0.01} = \frac{2\pi}{\lambda} \rightarrow \lambda = \pi/50$

$v = \omega/k$

(c) $f = ?$ $\frac{1}{0.05} = 2\pi f \rightarrow f = \frac{1}{0.1\pi}$

wave speed $\rightarrow \frac{1/0.01}{1/0.05} = 5$

To find phase constant at 2 different points

$\Delta\phi = 2\pi \frac{\Delta x}{\lambda}$

same disp, same speed, same relative pos θ

$y = 2 \sin\left(\frac{\pi}{2}(2t-5x)\right) \rightarrow v_p = 2\pi \cos\left(\frac{\pi}{2}(2t-5x)\right)$

$V_{max} = \pm A\omega$

what is particle speed?

(a) at origin at $t=0.5 \rightarrow x=0$

(b) at origin at $t=1.5 \rightarrow x=0$

(c) at origin at $t=6.5, x=2\pi$

To find max value of velocity $= 2\pi A \cos$ is b/w $(1 \text{ and } -1)$

$v = \omega/k$

$v_p = -v \left(\frac{1}{5}\right)$ Proof

$\rightarrow$ slope of y/x

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Speed of wave on string :- [Transverse wave]

Speed $\propto \frac{1}{\text{mass of string}}$

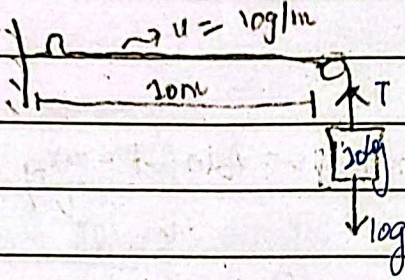
Is heavy wave/string takes more time

Speed $\propto$ Tension

$$v = \sqrt{\frac{T}{\mu}}$$

$$\mu = m/L \text{ [linear mass density]}$$

Q)



$$v = \sqrt{\frac{T}{\mu}} \rightarrow \sqrt{\frac{100}{10 \times 10^{-3}}}$$

$$v = T\omega$$

$$t = \frac{2}{v} \rightarrow \approx 0.6$$

$$\text{Speed} = \sqrt{xg}$$

$$\int \frac{dx}{\sqrt{x}} = \int \sqrt{g} dt$$

$$t_{\text{motion}} = 2 \sqrt{\frac{L}{g}}$$

wave on string g.s.

ENERGY of wave :-

$$\text{work done by tension} = P \cdot \epsilon = \Delta u = T (du - dx)$$

$$P \cdot \epsilon = \frac{du}{dx} = \frac{T}{2} \left(\frac{dy}{dx} \right)^2$$

$$y = A \sin(\omega t - kx)$$

$$\frac{dy}{dx} = \pm k A \cos(\omega t - kx)$$

$$P \cdot \epsilon = \frac{\mu A^2 \omega^2}{2} \cos^2(\omega t - kx)$$

$$T \cdot \epsilon = \mu A^2 \omega^2 \cos^2(\omega t - kx)$$

$$\text{instantaneous power} = A^2 \omega^2 v^2 \cos^2(\omega t - kx)$$

$$\text{avg power} \rightarrow \frac{1}{2} P_{\text{avg}} = \frac{A^2 \omega^2 v}{2}$$

$$\text{Intensity} = \frac{P}{A} = \frac{A^2 \omega^2 v}{2A}$$

2 cycle diagram line

$$\text{Intensity} = 2\pi A^2 f^2 \rho v$$

avg intensity

Amplitude

density of medium

speed

GENIUS

Interference of wave :-

Superposition principle = $y_{net} = \vec{y}_1 + \vec{y}_2$

① $A_1 = A \sin(\omega t + kx)$

→ So add both eq. } become Resultant

② $y_2 = A_2 \sin(\omega t + kx)$

frequencies remain same. $k = 2\pi/\lambda$, $\omega = 2\pi f$, $v = f\lambda$ $f, \omega, v, \lambda \rightarrow$ same

$$y_{net} = A_{net} \sin(\omega t + \theta)$$

$$\phi = kx_2 - kx_1 \rightarrow \text{Phase} = \frac{2\pi}{\lambda} \Delta x$$

$$y_{net} = [2y_m \cos(\phi/2)] \sin(\omega t - kx + \phi/2)$$

$$\tan \theta = \frac{A_2 \sin \theta}{A_1 + A_2 \cos \theta}$$

$$I = kA^2$$

Constructive ✓

Destructive

$$\text{Speed of sound} = \sqrt{\frac{\gamma}{\rho}}$$

$$A_{net} / I_{net} \rightarrow \text{max}$$

$$A_{net} \rightarrow \text{min}$$

$$\cos \phi \rightarrow \text{max} \rightarrow 1$$

$$\cos \phi \rightarrow \text{min} \rightarrow -1$$

$$A_{net} = (A_1 + A_2)^2$$

$$I_{net} = 4I$$

$$A_{net} = (A_1 - A_2)^2$$

$$I_{net} = 0$$

$$v = \sqrt{\frac{\gamma R T}{M}}$$

$$I_{net} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{net} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$A_{net} = 0$$

$$\phi = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\phi = (2n+1)\pi$$

$$m_2 \propto m \cos \theta$$

"Resultant waves have larger $\theta = n\pi$ ""multiples of π radian" $(2n+1)\frac{\pi}{2} = 2n\pi + \pi$ f_n amplitude when either wave " $\phi = 2n\pi$

$$f_n = nfv$$

$$L_n = \frac{2L}{n}$$

$$f_n = \frac{n v}{2L}$$

$$\lambda_n = \frac{\lambda_1}{n}$$

Intermediate $\rightarrow$ b/w 0 and 180°

"Amplitude reduces to 0"